Gaussian Mixture Model Clustering with Incomplete Data

From paper reproduction to a one-stage view of imputation + clustering

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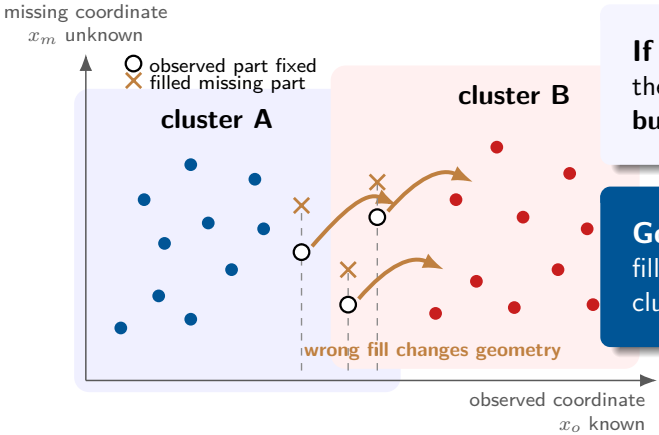
Roadmap

1. Motivation
2. Formulation
3. Algorithm
4. Implementation
5. Results
6. Takeaways

Missing values do not just remove numbers.

They bend the cluster geometry.



If we impute first
the filled values may look plausible
but destroy separation.

Goal
fill the blanks in a way that helps clustering.

The Paper in One Sentence

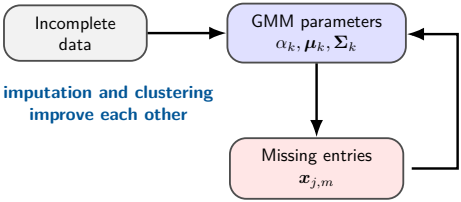
Do not separate imputation and clustering.

Optimize them together inside one Gaussian mixture model.

Two-stage

imputation never sees the clustering objective

One-stage



imputation and clustering improve each other

Notation Before the Equations

Data with missing coordinates

$$\mathbf{X} \in \mathbb{R}^{n \times d}, \qquad \boldsymbol{x}_j \in \mathbb{R}^d, \qquad \boldsymbol{x}_j = \begin{bmatrix} \boldsymbol{x}_{j,o} \\ \boldsymbol{x}_{j,m} \end{bmatrix}$$

- $\mathcal{O}_j$: observed feature indices
- $\mathcal{M}_j$: missing feature indices
- $\boldsymbol{x}_{j,o}$ is locked by observation
- $\boldsymbol{x}_{j,m}$ is optimized by the algorithm

GMM quantities

K	number of Gaussian components
α_k	mixture weight of component k
$\boldsymbol{\mu}_k$	component mean vector
$\boldsymbol{\Sigma}_k$	component covariance matrix
$\boldsymbol{\Lambda}_k = \boldsymbol{\Sigma}_k^{-1}$	component precision matrix
γ_{jk}	posterior responsibility for sample j
P_{jk}	unnormalized component score
$\boldsymbol{\mu}_{k,o}, \boldsymbol{\Lambda}_{k,mm}$	observed/missing blocks induced by $\mathcal{O}_j, \mathcal{M}_j$

Joint Optimization View

$$\max_{\mathbf{X},\{\alpha_k,\boldsymbol{\mu}_k,\boldsymbol{\Sigma}_k\}_{k=1}^{K}} \sum_{j=1}^{n} \log\left(\sum_{k=1}^{K} \alpha_k \mathcal{N}(\boldsymbol{x}_j \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)\right)$$

$$\text{s.t. } \boldsymbol{x}_{j,o} = \boldsymbol{x}_{j,o}^{\text{obs}}, \qquad \boldsymbol{x}_j = \begin{bmatrix} \boldsymbol{x}_{j,o} \\ \boldsymbol{x}_{j,m} \end{bmatrix}$$

The optimization variable is the completed matrix $\mathbf{X}$,
not only the GMM parameters.

Two hidden layers

- γ_{jk}: cluster uncertainty
- $\boldsymbol{x}_{j,m}$: feature uncertainty

Observed entries are locked.
Only the missing block moves.

The Three Equations That Matter

E-step

$$\gamma_{jk} = \frac{\alpha_k \mathcal{N}(\boldsymbol{x}_j \mid \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k)}{\sum_{\ell=1}^{K} \alpha_\ell \mathcal{N}(\boldsymbol{x}_j \mid \boldsymbol{\mu}_\ell, \boldsymbol{\Sigma}_\ell)}$$

soft cluster membership

M-step params

$$N_k = \sum_{j=1}^{n} \gamma_{jk}, \qquad \alpha_k = \frac{N_k}{n}$$

$$\boldsymbol{\mu}_k = \frac{1}{N_k} \sum_{j=1}^{n} \gamma_{jk} \boldsymbol{x}_j$$

$$\boldsymbol{\Sigma}_k = \tfrac{1}{N_k} \textstyle\sum_{j=1}^{n} \gamma_{jk} (\boldsymbol{x}_j - \boldsymbol{\mu}_k)(\boldsymbol{x}_j - \boldsymbol{\mu}_k)^\top$$

M-step data

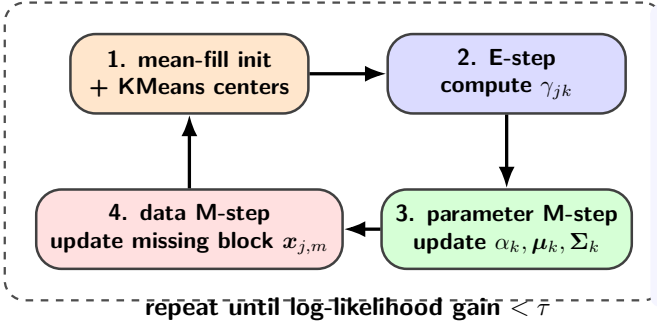
$$\boldsymbol{x}^\star_{j,m} = \mathbf{A}_j^{-1} \boldsymbol{b}_j$$

$$\mathbf{A}_j = \sum_{k=1}^{K} P_{jk} \boldsymbol{\Lambda}_{k,mm}$$

$$\boldsymbol{b}_j = \sum_{k=1}^{K} P_{jk} \Big(\boldsymbol{\Lambda}_{k,mo} \boldsymbol{\mu}_{k,o} + \boldsymbol{\Lambda}_{k,mm} \boldsymbol{\mu}_{k,m} - \boldsymbol{\Lambda}_{k,mo} \boldsymbol{x}_{j,o} \Big)$$

precision-weighted
conditional reconstruction

Alternating Optimization Loop



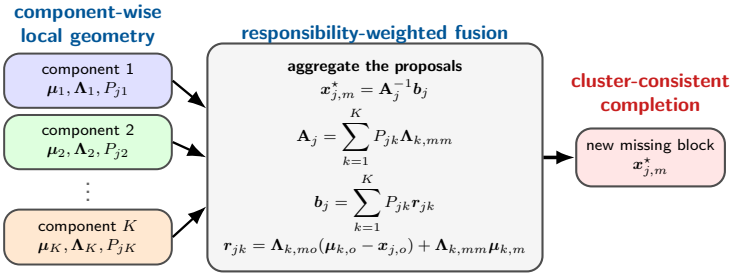
Why it works

- every block update pushes the objective upward
- EM handles soft assignments
- closed-form $\boldsymbol{x}_{j,m}$ update refreshes the surrogate data matrix

Guarantee: monotone improvement
Reality: local optimum

Why the Missing-Data Update Is the Real Star

Each Gaussian component proposes a local completion.
Responsibilities decide how much each proposal matters.



Not a generic imputer. It is a **cluster-aware geometric reconstruction**.

How the Repository Mirrors the Math

`src/gmm_missing.py`
alternating estimator

`notebooks/step_by_step_experiment.ipynb`
single-cycle explanation

`notebooks/GMM_Nhat.ipynb`
CC GENERAL reproduction

`src/airplane_crashes_pipeline.py`
real-data workflow

report & slides
formulas
figures
artifacts
discussion

Core file

`src/gmm_missing.py`

Numerical safeguards

- mean fill only for initialization
- KMeans initialization
- covariance ridge regularization
- log-sum-exp in E-step
- fallback when precision blocks are singular

Code-to-Equation Mapping

Equation block	Meaning	In code
γ_{jk}	posterior responsibilities	`_e_step()` with log-domain Gaussian scores
$\alpha_k, \boldsymbol{\mu}_k, \boldsymbol{\Sigma}_k$	standard GMM M-step	`_m_step_params()`
$\boldsymbol{x}^{\star}_{j,m}$	precision-weighted update for missing entries	`_m_step_data()`
fit loop	alternating ascent until tolerance	`fit()` / `fit_scale()`

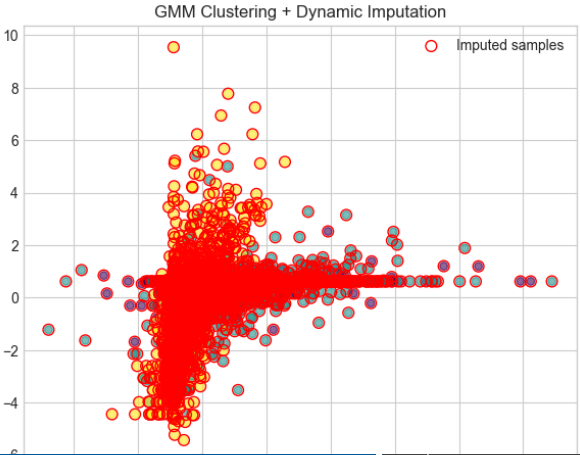
The implementation is not a black box.
The paper equations are visible almost line by line.

Local Reproduction: CC GENERAL

Setup

- 8,950 customers
- 17 numeric features
- injected missingness $\approx 30\%$
- KMeans init + alternating GMM updates

Observed run
convergence after **199** iterations
silhouette score **0.1489**

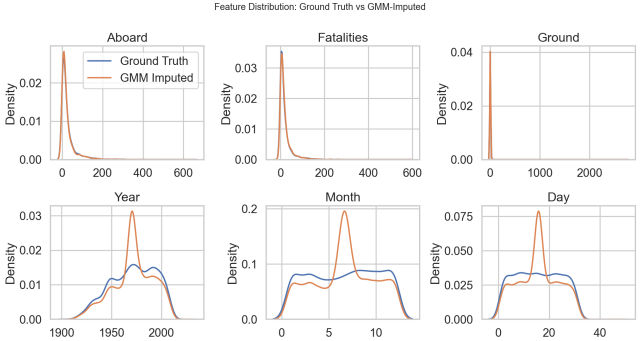


Application Case: Airplane Crashes

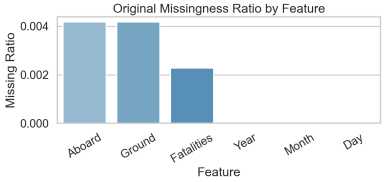
Pipeline

- 5,236 usable rows
- 6 numeric features
- MCAR masking at 20%
- $K = 4$ Gaussian components

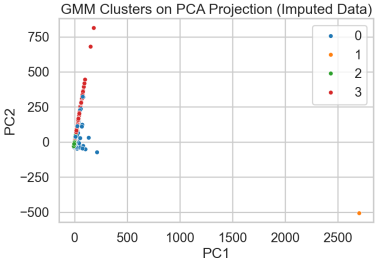
RMSE on masked entries
20.0588



What the Visual Artifacts Say



feature-level missingness profile



geometry after dynamic imputation

The method is not only mathematically elegant.
It survives contact with real, dirty tabular data.

Strengths, Limits, and Research Headroom

Strengths

- one-stage objective
- cluster-aware imputation
- full covariance geometry
- monotone ascent behavior

Limits

- local optimum only
- cubic cost in dimension
- sensitive to K, init, regularization
- assumes synthetic missingness is informative enough

Next

- diagonal / tied / sparse covariance
- automatic model selection
- MAR/MNAR-aware extensions
- uncertainty-preserving imputation

Take-home message

For incomplete-data clustering,

the best imputation is the one optimized for the clustering objective.

Gaussian geometry + alternating optimization + precision-based reconstruction

Q & A